

Chapter 5 Present Worth (Value) Analysis

- **Introduction**

- Given a set of “feasible” alternatives, engineering economy attempts to identify the “best” (most viable) alternative(s) from an economic perspective.
- Economic perspective requires a quantitative criteria for decision making.
- In this chapter, we study the *present worth* criteria.

- **Types of economic projects**

- *Mutually exclusive alternatives*
 - From a set of feasible alternatives, pick *only* one. E.g., which car to buy.
 - Mutually exclusive alternatives “compete” with each other.
- *Independent projects*
 - From a set of feasible alternatives, select as many as possible to meet the economic criteria the most. E.g., where to invest money?
 - In the absence of a budget constraint, choose all alternatives that do better than the “do nothing” alternative.

- **Do nothing (status-quo) alternative**
 - This is the alternative of not changing the current situation.
E.g., keep money in a saving account, rather than in stocks.
- **Cash flow types for projects**
 - *Revenue* – each alternative generates costs and revenues over the life of the project. E.g., what product to introduce?
 - Criteria: Select the alternative that maximizes the economic measure of merit, which is profit-based.
 - *Service* – each alternative has only cost cash flows. Revenues are the same for all alternatives. E.g., which 100-seat plane to buy?
 - Criteria: Select the alternative that minimizes the economic measure of merit, which is cost-based.
- **Present Worth (PW) analysis**
 - This is the process of obtaining the equivalent worth of future cash flows at present time (time 0).
 - That is, finding PW of cash flows.
 - We say that future cash flows are “discounted” to time 0.
 - For revenue-type projects, the higher the PW, the better off.
 - PW is evaluate based on an interest rate, which is equal to the organization’s MARR.

- **PW analysis of equal-life alternatives**

- *Mutually exclusive projects*

- For one project, it is financially viable if $PW \geq 0$.
 - For two or more alternatives, select the one with the (numerically) largest PW value, provided $PW \geq 0$ for this alternative.

- *Independent Projects*

- Select *all* projects with $PW \geq 0$
 - However, in practice a budget limit exists (see Ch. 12)

- **PW analysis of different-life alternatives**

- For alternatives with unequal lives the rule is

- PW must be compared over the same number of years.*

- This is called “equal service” requirement

- Equal service requirement can be met in two ways

- *LCM*

- Evaluate alternatives over the lowest common multiple of lives. E.g., lives of 4 and 6, use $n = 12$.
 - Assume reinvestment at same cash flow estimates in each life cycle of the LCM planning horizon.

- *Study period*

- Assume a fixed planning horizon and evaluate the alternatives over it.
 - Ignore cash flows beyond the planning horizon.

- **LCM assumptions**

- The service provided is needed for LCM years or more.
- The selected alternative is *repeated* over each the life cycle of the LCM in exactly the same manner.
- Cash flow estimates are the same in every life cycle.

- **Study period and alternative life**

- Depending on the life of an alternative, three cases could occur when adopting the study period approach.
 1. *Alternative life equal to the study period.* No adjustment to the cash flow is required.
 2. *Alternative life longer than the study period.* An implied salvage value must be added to the alternative at the end of the study period.
 - The salvage value may be estimated based on the *market value* of the asset generating the cash flows.
 - It may be also based on the PW of remaining cash flows.
 3. *Alternative life shorter than the study period.* Assumptions must be made on what happens in the additional years between end of life and end of study period.
 - For service (cost) alternatives, one can estimate the costs of continuing service over the additional years.

- For revenue alternatives, one may assume that the net receipts are invested at MARR for the additional years.

- **Future worth (FW) analysis**

- Similar to PW analysis but uses future instead of present values. (MARR is also used to find future values.)
- Utilized when
 - A prime goal is to maximize future wealth of stockholders.
 - Asset may be sold after some time of startup (e.g., buy a company and sell it in three years).
 - Projects will not “come online” until end of investment period (e.g., construction projects).
- FW and PW criteria are equivalent in comparing alternatives.

- **Capitalized Cost (CC) analysis**

- Capitalized cost is the present worth of a project that lasts forever.
- This occurs
 - Public Sector Projects. E.g., roads, bridges, dam.
 - Not-for-profit organization endowments.
- For these projects, the life cycle, n , is either very long, indefinite, or infinity.
- The CC for an infinite uniform series of cash flows (with annuity A) is

$$CC = \lim_{n \rightarrow \infty} \frac{A}{i} \left[1 - \frac{1}{(1+i)^n} \right] = \frac{A}{i}.$$

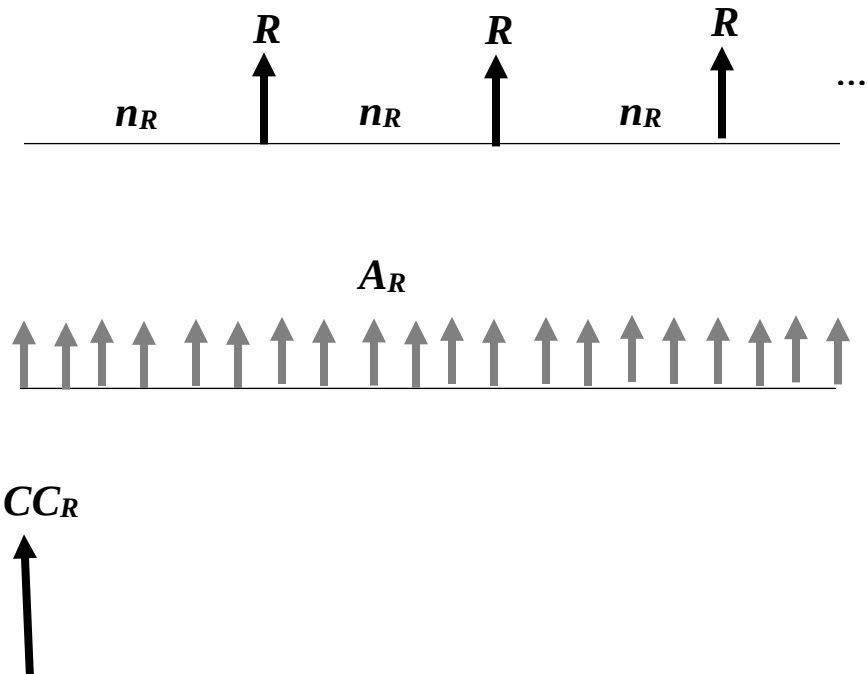
➤ To evaluate CC for any cash flow, do the following.

- For nonrecurring (one-time only) cash flows
 - The CC of the cash flows is their PW.
- For a recurring cash flow of value R , that repeats every n_R years
 - Find equivalent uniform annual worth through one life cycle of recurring amounts,

$$A_R = R(A/F, i, n_R) = Ri / [(1+i)^n - 1].$$

- Find equivalent CC for the AR series,

$$CC_R = A_R / i = R / [(1+i)^n - 1].$$



- Alternatives that have infinite lives can be compared on the basis of CC, which is equivalent to PW criteria.
- When an alternative with a finite life is to be compared with another having infinite life, the guidelines for alternatives with life shorter than study period (here infinity) are applied.

- **Payback period analysis**

- Payback period is the estimated time it will take for the revenues of a project to recover the initial investment.
- The payback period, n_p , is such that

$$0 = -P + \sum_{t=1}^{n_p} NCF_t(P/F, i, t),$$

where P is the initial investment and NCF_t is the net cash flow at time t .

- This equation can be solved using trial and error or using a computer package (e.g., Excel solver.)
- If $i = 0\%$, $0 = -P + \sum_{t=1}^{n_p} NCF_t$. If, in addition, $NCF_t = NCF$ for all t , $n_p = P/NCF$.
- This method estimate of n_p is often used in practice for quick initial screening.
- Payback period analysis should not be used as the primary means of making an accept/reject decision on an alternative.
- E.g., one reason for caution with payback analysis is that it ignores cash flows after time n_p .

- **Payback period analysis in Excel**

- For a project of life n years, having an initial investment, P , at Time 0, a uniform cash flow, A , from Times 1 to n , and a terminal value, at time n , F , a built-in function, gives the payback period, $\text{NPER}(i,A,P,F)$.
- If the cash flows don't follow the (A, P, F) structure, then typically trial and error is needed to find the payback period. See the example in the Excel file on the course website.